

Francis Njoroge Wanjiru

Nairobi, Kenya | francisnjorogedev@gmail.com | +254 724 015 861 | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

FULL STACK SOFTWARE ENGINEER

React • TypeScript • AI Applications • Conversational AI • Twilio • REST APIs

Professional Summary

Full Stack Software Engineer with 4+ years of experience designing and developing scalable web applications, enterprise business systems, and AI-powered conversational solutions using React, Next.js, TypeScript, Node.js, Python, GraphQL, REST APIs, and PostgreSQL. Experienced in developing WhatsApp chatbots with Twilio, integrating LLM-powered AI capabilities into customer-facing applications, and delivering software across the full development lifecycle. Skilled in AI-assisted development using GitHub Copilot, Claude, Cursor AI, and ChatGPT to improve development speed, code quality, and delivery.

Core Skills

Frontend: React, TypeScript, Next.js, JavaScript (ES6+), Redux, Zustand, HTML5, CSS3, Responsive Web Design, Bootstrap, Material UI

Backend: Python (Flask, Django), Node.js, Express, GraphQL, REST APIs, JWT Authentication & Role-Based Access Control (RBAC)

Databases: PostgreSQL, MySQL, SQLite, MongoDB

Conversational AI & AI Development: Twilio WhatsApp API, WhatsApp Chatbots, AI Assistants, LLM Integration, Prompt Engineering, AI-powered Workflow Automation, GitHub Copilot, Claude, Cursor AI, ChatGPT

DevOps: Docker, Git, GitHub Actions, Linux, CI/CD

Professional Experience

Full Stack Product Engineer | Product Prototyping & Digital Innovation

Kafew Digital Solutions | Nairobi

July 2024 – Present

- Designed and developed scalable full-stack applications using React, TypeScript, Express.js, Flask, and PostgreSQL.
- Built reusable React component libraries and scalable frontend architectures, reducing development effort across multiple projects using React, Redux, Zustand, and TypeScript.
- Developed GraphQL and REST API integrations enabling secure, real-time communication between frontend applications and enterprise backend services.
- Engineered conversational AI solutions using Twilio WhatsApp APIs for customer engagement and workflow automation.
- Optimized application performance through efficient state management, API design, and responsive frontend implementation.
- Implemented Docker-based deployment workflows and GitHub Actions to support continuous integration and automated application delivery.

- Developed secure integrations with third-party services using REST APIs, GraphQL, and webhooks, enabling real-time communication between enterprise systems and conversational AI platforms.
- Integrated LLM-powered AI capabilities into customer-facing applications, automating workflows and improving user experience.
- Leveraged GitHub Copilot, Claude, Cursor AI, and ChatGPT to accelerate development, automate repetitive coding tasks, improve testing, and enhance code quality.
- Collaborated with stakeholders through the complete software development lifecycle, from requirements gathering to deployment and production support.

Selected AI, Conversational Solutions & Enterprise Applications

SheCare; Women's Self-Care AI Assistant(Team Project)

React | Flask | Twilio | OpenAI API | PostgreSQL | REST APIs

- Developed an AI-powered WhatsApp healthcare assistant using Twilio for conversational interactions.
- Implemented symptom guidance, prescription interpretation, and geolocation services to help users locate nearby healthcare facilities.
- Integrated LLM capabilities to deliver contextual health information while maintaining an intuitive conversational experience.

WhatsApp CRM for Beauty Salon

React | Node.js | Express | Twilio | PostgreSQL | GraphQL | REST APIs

GitHub: <https://github.com/Ka-few/whatsapp-business-os>

- Built an intelligent WhatsApp CRM platform that automates appointment booking, customer engagement, reminders, and service inquiries.
- Integrated Twilio messaging APIs with backend business workflows to streamline salon operations.
- Designed the solution as a multi-tenant SaaS platform capable of supporting multiple businesses.

ShambaSmart FARM Management Application

Built an offline-first farm management platform featuring responsive dashboards, financial reporting, operational analytics, and data visualization using React, TypeScript, SQLite, and REST APIs.

React | TypeScript | SQLite | REST APIs | GraphQL | Tauri

Beauty Salon System

Developed a salon management system with appointment scheduling, customer management, authentication, responsive user interfaces, and business reporting using React, Flask and REST APIs.

React | Flask | PostgreSQL | REST APIs

Multi-Branch Retail & POS Management System

Designed a retail management platform supporting POS transactions, inventory management, customer credit tracking, reporting dashboards, and role-based access control.

React | TypeScript | Node.js | Express | SQLite | Tauri | REST APIs

Sales & IT Support Officer

Datacomm Systems & Solutions | Nakuru

Nov 2016 – Apr 2024

- Delivered L1/L2 technical support for hardware, software, and network incidents.
- Installed and maintained client IT systems, routers, and network infrastructure.
- Supported enterprise users across deployment, troubleshooting, and operational IT workflows.

Education

Bachelor of Science in Information Technology

Jomo Kenyatta University of Agriculture & Technology | 2013

Certifications & Professional Development

Software Engineering Certificate

Moringa School | May 2025 – Nov 2025

- Intensive training in full-stack development, system design, microservices architecture, and modern DevOps practices.
- Built scalable backend systems and production-ready applications.

Oracle Database: Introduction to SQL

Computer Pride | Jan 2016 – May 2016

Oracle Database 11g: Administration Workshop I

Computer Pride | Jan 2016 – May 2016

- Database installation, user management, backup and recovery concepts, performance monitoring, and system optimization.